

Technical Questions

1. Who is known as the father of C language?

- A) Bjarne Stroustrup
- B) Dennis Ritchie
- C) James Gosling
- D) Dr. E.F. Codd

Answer: B) Dennis Ritchie

2. Which of the following is not a valid C variable name?

- A) int number;
- B) float rate;
- C) int variable_count;
- D) int \$main;

Answer: D) int \$main;

3. What is the size of `int` data type in a 32-bit compiler?

- A) 2 bytes
- B) 4 bytes
- C) 8 bytes
- D) 1 byte

Answer: B) 4 bytes

4. Which of the following is a logical operator?

- A) &
- B) &&
- C) ||
- D) Both B and C

Answer: D) Both B and C

5. The process of finding errors in software code is called:

- A) Compiling
- B) Testing
- C) Debugging
- D) Running

Answer: C) Debugging

6. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Tree
- D) Graph

Answer: B) Stack

7. Which data structure uses FIFO (First In First Out) order?

- A) Stack
- B) Queue
- C) Linked List
- D) Array

Answer: B) Queue

8. In Object-Oriented Programming, wrapping data and functions into a single unit is called:

- A) Inheritance
- B) Polymorphism
- C) Encapsulation
- D) Abstraction

Answer: C) Encapsulation

9. Which concept allows a class to derive properties from another class?

- A) Polymorphism
- B) Inheritance
- C) Encapsulation

D) Overloading

Answer: B) Inheritance

10. What is the time complexity of binary search?

A) $O(n)$

B) $O(n^2)$

C) $O(\log n)$

D) $O(1)$

Answer: C) $O(\log n)$

11. Which sorting algorithm has the best average-case time complexity?

A) Bubble Sort

B) Selection Sort

C) Merge Sort

D) Insertion Sort

Answer: C) Merge Sort ($O(n \log n)$)

12. What does OS stand for?

A) Open Source

B) Operating System

C) Optical Sensor

D) Output System

Answer: B) Operating System

13. Which of the following is NOT an Operating System?

A) Windows

B) Linux

C) Oracle

D) Android

Answer: C) Oracle (It is a Database)

14. What is the main function of the ALU?

- A) Store data
- B) Perform arithmetic and logical operations
- C) Control computer units
- D) Monitor output

Answer: B) Perform arithmetic and logical operations

15. IP address stands for:

- A) Internet Provider
- B) Internet Protocol
- C) Internet Procedure
- D) Intranet Protocol

Answer: B) Internet Protocol

16. Which layer of the OSI model is responsible for routing?

- A) Physical Layer
- B) Data Link Layer
- C) Network Layer
- D) Transport Layer

Answer: C) Network Layer

17. HTTP stands for:

- A) Hyper Text Transfer Protocol
- B) High Transfer Text Protocol
- C) Hyper Text Transmission Protocol
- D) Hyper Transfer Text Protocol

Answer: A) Hyper Text Transfer Protocol

18. In C, strings are terminated by:

- A) '.'
- B) '\0' (Null character)

C) '\n'

D) '0'

Answer: B) '\0' (Null character)

19. Which keyword is used to define a constant in C?

A) constant

B) const

C) define

D) static

Answer: B) const

20. Which of the following is a non-linear data structure?

A) Array

B) Linked List

C) Stack

D) Tree

Answer: D) Tree

21. The pointer which points to nothing is called:

A) Void Pointer

B) Null Pointer

C) Dangling Pointer

D) Wild Pointer

Answer: B) Null Pointer

22. A constructor in C++:

A) Has a return type

B) Has the same name as the class

C) Cannot be overloaded

D) Is called manually

Answer: B) Has the same name as the class

23. Java is a:

- A) Platform Dependent language
- B) Platform Independent language
- C) Low-level language
- D) Sequence-oriented language

Answer: B) Platform Independent language

24. Which of the following is not a Java keyword?

- A) static
- B) try
- C) Integer
- D) new

Answer: C) Integer (It is a Wrapper class, not a keyword)

25. In a Relational Database, a table is also known as:

- A) Tuple
- B) Relation
- C) Attribute
- D) Entity

Answer: B) Relation

26. SQL stands for:

- A) Structured Query Language
- B) Standard Query Language
- C) Simple Query Language
- D) Sequential Query Language

Answer: A) Structured Query Language

27. Which command is used to create a table in SQL?

- A) MAKE TABLE

B) CREATE TABLE

C) ADD TABLE

D) NEW TABLE

Answer: B) CREATE TABLE

28. The primary key of a table must be:

A) Unique

B) Not Null

C) Both Unique and Not Null

D) Text based

Answer: C) Both Unique and Not Null

29. What is a Deadlock in OS?

A) System crash

B) Two processes waiting for each other indefinitely

C) A fast process

D) Memory leak

Answer: B) Two processes waiting for each other indefinitely

30. Virtual Memory is:

A) A large secondary memory

B) An illusion of large main memory

C) A type of supercomputer memory

D) Cache memory

Answer: B) An illusion of large main memory

31. Which scheduler selects processes from the disk and loads them into memory?

A) Short-term scheduler

B) Long-term scheduler

C) Medium-term scheduler

D) Dispatcher

Answer: B) Long-term scheduler

32. DNS stands for:

- A) Domain Name System
- B) Data Network Service
- C) Data Name System
- D) Domain Network Service

Answer: A) Domain Name System

33. Which protocol is used to send email?

- A) FTP
- B) SMTP
- C) HTTP
- D) POP3

Answer: B) SMTP

34. 1 Byte is equal to:

- A) 4 bits
- B) 8 bits
- C) 16 bits
- D) 32 bits

Answer: B) 8 bits

35. The binary equivalent of decimal 10 is:

- A) 1000
- B) 1010
- C) 1100
- D) 1001

Answer: B) 1010

36. Which of the following is a volatile memory?

- A) ROM
- B) RAM
- C) Hard Disk
- D) Flash Drive

Answer: B) RAM

37. A linked list is best described as:

- A) Random access structure
- B) Sequential access structure
- C) Index based structure
- D) Static structure

Answer: B) Sequential access structure

38. The worst-case complexity of Quick Sort is:

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

Answer: C) $O(n^2)$

39. Which OOP principle allows the same function name to be used for different types of operations?

- A) Inheritance
- B) Polymorphism
- C) Encapsulation
- D) Abstraction

Answer: B) Polymorphism

40. `malloc()` in C returns a pointer of type:

- A) `int *`
- B) `char *`

C) void *

D) null

Answer: C) void *

41. In C++, `cin` and `cout` are:

A) Functions

B) Classes

C) Objects

D) Operators

Answer: C) Objects

42. Which is not a type of inheritance?

A) Single

B) Multiple

C) Distributed

D) Hierarchical

Answer: C) Distributed

43. Which of the following is used to speed up database retrieval?

A) Trigger

B) Index

C) View

D) Stored Procedure

Answer: B) Index

44. DML stands for:

A) Data Manipulation Language

B) Data Modeling Language

C) Data Management Language

D) Data Mining Language

Answer: A) Data Manipulation Language

45. Which join returns all records when there is a match in either left or right table?

- A) Inner Join
- B) Left Join
- C) Right Join
- D) Full Outer Join

Answer: D) Full Outer Join

46. "Paging" is a memory management scheme that eliminates the need for:

- A) Contiguous allocation of physical memory
- B) Virtual Memory
- C) Segmentation
- D) Partitioning

Answer: A) Contiguous allocation of physical memory

47. Context switching is part of:

- A) Interrupt handling
- B) Process scheduling
- C) Memory management
- D) File management

Answer: B) Process scheduling

48. A firewall is used for:

- A) Virus protection
- B) Security against unauthorized access
- C) Data backup
- D) Cooling the system

Answer: B) Security against unauthorized access

49. IPv4 addresses are:

- A) 16 bits

- B) 32 bits
- C) 64 bits
- D) 128 bits

Answer: B) 32 bits

50. HTML stands for:

- A) Hyper Text Markup Language
- B) High Text Marking Language
- C) Hyper Text Marking Language
- D) Hyper Tool Markup Language

Answer: A) Hyper Text Markup Language

51. What is the full form of URL?

- A) Uniform Resource Locator
- B) Universal Resource Locator
- C) Uniform Reference Locator
- D) Universal Reference Locator

Answer: A) Uniform Resource Locator

52. In C, array index starts from:

- A) 1
- B) 0
- C) -1
- D) User defined

Answer: B) 0

53. Which loop is guaranteed to execute at least once?

- A) for
- B) while
- C) do-while
- D) None

Answer: C) do-while

54. The keyword used to transfer control from a function back to the calling function is:

- A) switch
- B) goto
- C) go back
- D) return

Answer: D) return

55. `int arr[5];` - How many elements does this array hold?

- A) 4
- B) 5
- C) 6
- D) Unknown

Answer: B) 5

56. Which of the following is not a primitive data type in Java?

- A) int
- B) float
- C) String
- D) boolean

Answer: C) String (It is a class)

57. `try-catch` block is used for:

- A) Exception Handling
- B) String Manipulation
- C) File Handling
- D) Looping

Answer: A) Exception Handling

58. What is the purpose of the `final` keyword in Java?

- A) To make a variable constant
- B) To prevent method overriding
- C) To prevent inheritance
- D) All of the above

Answer: D) All of the above

59. Normalization is a process of:

- A) Removing errors
- B) Organizing data to minimize redundancy
- C) Sorting data
- D) Encrypting data

Answer: B) Organizing data to minimize redundancy

60. ACID properties in DBMS stand for:

- A) Atomicity, Consistency, Isolation, Durability
- B) Accuracy, Consistency, Isolation, Durability
- C) Atomicity, Concurrency, Isolation, Durability
- D) Access, Consistency, Isolation, Database

Answer: A) Atomicity, Consistency, Isolation, Durability

61. Which scheduling algorithm suffers from starvation?

- A) FCFS
- B) Round Robin
- C) Priority Scheduling
- D) SJF

Answer: C) Priority Scheduling (Low priority processes may wait indefinitely)

62. The core of the Linux operating system is called:

- A) Shell
- B) Kernel
- C) Terminal

D) Command Prompt

Answer: B) Kernel

63. Which protocol provides reliable delivery of data packets?

A) UDP

B) TCP

C) IP

D) ICMP

Answer: B) TCP

64. VPN stands for:

A) Virtual Public Network

B) Virtual Private Network

C) Viral Private Network

D) Virtual Personal Network

Answer: B) Virtual Private Network

65. What is Phishing?

A) A type of fish

B) An attempt to acquire sensitive information by masquerading as a trustworthy entity

C) Checking network connectivity

D) A virus

Answer: B) An attempt to acquire sensitive information by masquerading as a trustworthy entity

66. In C, '&' operator is used for:

A) Value at address

B) Address of a variable

C) Logical AND

D) Bitwise OR

Answer: B) Address of a variable

67. `sizeof()` is a:

- A) Function
- B) Variable
- C) Operator
- D) Keyword

Answer: C) Operator

68. A function calling itself is called:

- A) Iteration
- B) Recursion
- C) Inversion
- D) Looping

Answer: B) Recursion

69. Which data structure is best for implementing a priority queue?

- A) Stack
- B) Linked List
- C) Heap
- D) Array

Answer: C) Heap

70. Which searching algorithm requires the array to be sorted?

- A) Linear Search
- B) Binary Search
- C) Hashing
- D) BFS

Answer: B) Binary Search

71. `Abstract` class in Java cannot be:

- A) Inherited
- B) Instantiated

- C) Compiled
- D) Overloaded

Answer: B) Instantiated

72. An interface in Java contains:

- A) Concrete methods only
- B) Abstract methods and constants (mostly)
- C) Variables
- D) Constructors

Answer: B) Abstract methods and constants (mostly)

73. `DELETE` vs `TRUNCATE`: Which one can be rolled back?

- A) DELETE
- B) TRUNCATE
- C) Both
- D) Neither

Answer: A) DELETE (In many standard contexts, TRUNCATE is DDL and implicitly commits, DELETE is DML)

74. `GRANT` and `REVOKE` are commands of:

- A) DDL
- B) DML
- C) DCL
- D) TCL

Answer: C) DCL

75. What is a Semaphore in OS?

- A) A variable used for process synchronization
- B) A memory block
- C) A file system
- D) A hardware device

Answer: A) A variable used for process synchronization

76. Thrashing occurs when:

- A) CPU is idle
- B) System spends more time paging than executing
- C) Disk is full
- D) Deadlock occurs

Answer: B) System spends more time paging than executing

77. The layer responsible for end-to-end communication in OSI model is:

- A) Transport Layer
- B) Network Layer
- C) Data Link Layer
- D) Session Layer

Answer: A) Transport Layer

78. MAC address is a:

- A) Physical Address
- B) Logical Address
- C) Port Address
- D) Network Address

Answer: A) Physical Address

79. 1 KB (Kilobyte) is equal to:

- A) 1000 bytes
- B) 1024 bytes
- C) 1024 bits
- D) 8 bytes

Answer: B) 1024 bytes

80. ASCII stands for:

- A) American Standard Code for Information Interchange
- B) American Standard Code for International Interchange
- C) American Scientific Code for Information Interchange
- D) Asian Standard Code for Information Interchange

Answer: A) American Standard Code for Information Interchange

81. Which operator is used to access members of a structure using a pointer in C?

- A) .
- B) ->
- C) *
- D) &

Answer: B) ->

82. `static` variable in C is initialized:

- A) Every time the function is called
- B) Only once
- C) Manually
- D) Never

Answer: B) Only once

83. Which is used to insert an element at the end of a queue?

- A) Push
- B) Pop
- C) Enqueue
- D) Dequeue

Answer: C) Enqueue

84. DFS uses which data structure?

- A) Queue
- B) Stack
- C) Heap

D) Tree

Answer: B) Stack

85. Which access modifier makes a member accessible only within the same class?

A) Public

B) Private

C) Protected

D) Default

Answer: B) Private

86. `super` keyword in Java refers to:

A) Current object

B) Parent class object

C) Static method

D) Main method

Answer: B) Parent class object

87. A view in SQL is:

A) A virtual table

B) A real table

C) A backup

D) An index

Answer: A) A virtual table

88. `HAVING` clause is used with:

A) SELECT

B) WHERE

C) GROUP BY

D) UPDATE

Answer: C) GROUP BY

89. A bootstrap program is stored in:

- A) RAM
- B) ROM
- C) Hard Disk
- D) Cache

Answer: B) ROM

90. Fragmentation is associated with:

- A) Paging
- B) Segmentation
- C) Disk management
- D) All of the above

Answer: D) All of the above

91. Which device operates at the Physical layer?

- A) Router
- B) Switch
- C) Hub
- D) Bridge

Answer: C) Hub

92. LAN stands for:

- A) Local Area Network
- B) Large Area Network
- C) Local Access Network
- D) Long Area Network

Answer: A) Local Area Network

93. The extension of a C++ source file is:

- A) .c
- B) .cpp

C) .java

D) .obj

Answer: B) .cpp

94. What is the output of `printf("%d", 10/3);` in C?

A) 3.33

B) 3

C) 3.0

D) Error

Answer: B) 3 (Integer division)

95. In a BST (Binary Search Tree), the left child is always:

A) Greater than the root

B) Smaller than the root

C) Equal to the root

D) Random

Answer: B) Smaller than the root

96. Which traversal visits the root node first?

A) Inorder

B) Preorder

C) Postorder

D) Level order

Answer: B) Preorder

97. Multiple inheritance is not directly supported in:

A) C++

B) Java

C) Python

D) Perl

Answer: B) Java (Supported through interfaces only)

98. `String` in Java is:

- A) Mutable
- B) Immutable
- C) Primitive
- D) None of these

Answer: B) Immutable

99. Which SQL constraint ensures no duplicate values in a column?

- A) PRIMARY KEY
- B) UNIQUE
- C) CHECK
- D) FOREIGN KEY

Answer: B) UNIQUE

100. In OSI model, data encryption happens at:

- A) Application Layer
- B) Presentation Layer
- C) Session Layer
- D) Transport Layer

Answer: B) Presentation Layer